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L_RAG: Knowledge-Graph-Augmented Vietnamese Legal RAG

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Final Project Presentation

Outline

Why Vietnamese Legal QA Is Difficult

Legal requirements

- Answers must be grounded in authoritative statutes.
- Citations must identify the correct article, clause, and item.
- Current validity and amendment history must be respected.
- Many questions require evidence across documents.

Why standard RAG is insufficient

- Dense search may miss exact statutory terminology.
- Fixed chunks break document hierarchy.
- Semantic similarity ignores legal precedence.
- LLMs may hallucinate or cite expired provisions.

Research problem

How can semantic matching, exact lexical search, statutory structure, temporal validity, and reranking be combined in one reliable legal RAG pipeline?

L_RAG: Objectives

1. Build an end-to-end hybrid pipeline combining dense E5 retrieval, BM25, and Reciprocal Rank Fusion.
2. Construct an in-memory legal knowledge graph for hierarchy, reading order, validity, authority, and context expansion.
3. Integrate multilingual rerankers: mMiniLMv2, BGE-Reranker-v2-m3, Jina-Reranker-v2, and Qwen3-Reranker-0.6B.
4. Generate answers from selected evidence using `gpt-4o-mini`, with strict citation and abstention rules.
5. Construct a standardized benchmark of 500 Vietnamese legal QA pairs.

Target output

A natural-language answer supported by traceable legal chunks, provision-level citations, authority metadata, and validity context.

Main Contributions

Complete pipeline

Modular ingestion, structuring, indexing, hybrid retrieval, graph traversal, reranking, and generation.

Dynamic legal graph

Query-time validity overlays and precedence sorting without rebuilding the vector index.

Benchmarking stack

A 500-query benchmark and support for four multilingual reranking backends with GPU-safe sequential loading.

Core contribution: evidence that is semantically relevant, structurally complete, temporally valid, and traceable.

Retrieval-Augmented Generation

Standard RAG

1. Encode the user query.
2. Retrieve semantically similar chunks.
3. Add retrieved evidence to the LLM prompt.
4. Generate a grounded answer.

Domain limitation

- Semantic ambiguity lowers precision.
- Fixed-size chunks lose long-range structure.
- Self-reflection methods do not directly model legal authority or validity.

Standard RAG reduces unsupported generation, but legal QA needs explicit statutory structure.

Legal Information Retrieval

Properties of legal text

- Deep hierarchy: code, chapter, article, clause, item.
- High sensitivity to exceptions and exact wording.
- Formal precedence among legal instruments.
- Dynamic amendments, replacements, and repeals.

Implication for retrieval

- A topically similar chunk may be legally inapplicable.
- Local evidence may require parent or sibling context.
- Validity depends on both time and document relationships.
- Neural embeddings alone cannot encode every explicit dependency.

Design requirement

The retriever must combine text relevance with authority, validity, hierarchy, and provenance.

Hybrid Retrieval, Reranking, and GraphRAG

Dense + sparse

E5 captures paraphrases; BM25 captures exact article numbers, legal codes, and domain terms.

Rank fusion

RRF combines heterogeneous ranked lists without requiring directly comparable scores:

$$\mathrm{RRF}(d) = \sum_{m} \frac{1}{60 + r_m(d)}$$

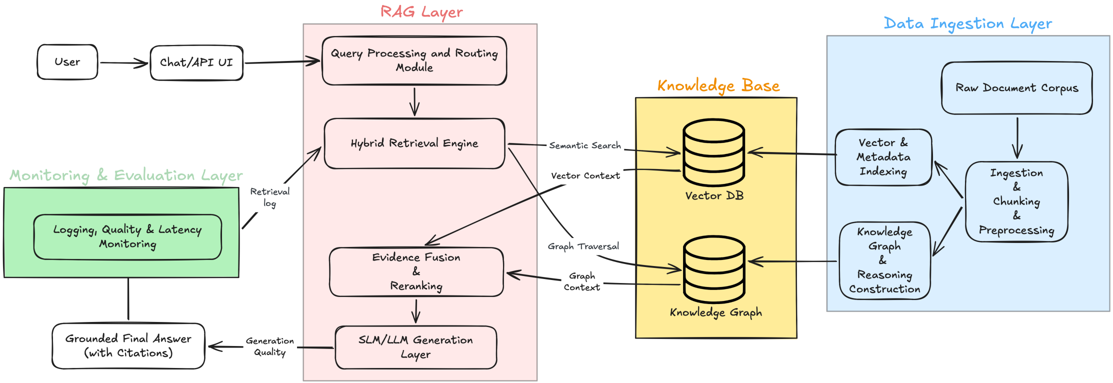
Graph + reranking

Graph traversal restores related legal context; cross-encoders jointly score each query–chunk pair.

L_RAG synthesis

Use dense and sparse retrieval for coverage, graph operations for legal coherence, and cross-encoders for final precision.

End-to-End L_RAG Architecture



Offline: corpus structuring, vector indexing, and graph construction. Online: hybrid retrieval, graph overlay, reranking, and cited generation.

Five-Stage Online Pipeline

User query + date/scope → Dense E5 + BM25 → Graph overlay + expansion → Fusion + reranking → Grounded LLM + citations

1. Retrieve a broad candidate pool from complementary dense and sparse indexes.
2. Apply legal whitelists and expand connected structural context.
3. Fuse and rerank candidates, then send the top 10 chunks to the generator.

Dense and Sparse Retrieval

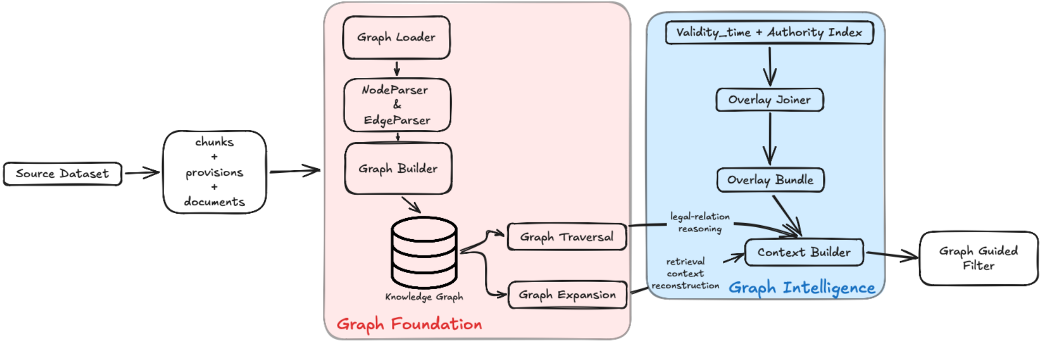
Dense vector search

- Model: `multilingual-e5-large` (560M, 1024 dimensions).
- FAISS `IndexFlatIP` over L2-normalized vectors.
- Embedding input combines document title, provision header, and chunk content.
- Strength: semantic paraphrases and natural-language questions.

Sparse keyword search

- BM25 with $k_1 = 1.5$ and $b = 0.75$.
- Vietnamese tokenization using `underthesea`.
- Strength: exact legal terms, document numbers, and citations.
- Dense and sparse lists are fused using RRF ($k = 60$).

Knowledge Graph Overlay



The graph coordinates document filtering, validity reasoning, precedence resolution, and reading-order context expansion.

Knowledge Graph Operations

Query-time reasoning

- Fold validity events relative to `as_of_date`.
- Resolve conflicts by legal authority and version.
- Produce `id_str_filter` whitelists for retrieval.
- Traverse only direction-verified legal edges.

Structural expansion

- `DOCUMENT_HAS_PROVISION`
- `PROVISION_HAS_CHUNK`
- `CHUNK_NEXT`
- `PROVISION_NEXT`

Sibling chunks are expanded in reading order to restore missing local context.

Implementation status

The in-memory loader, builder, traversal, expansion, overlay, and context modules pass all 27 knowledge-graph tests.

Sequential Multi-Reranker Stage

Supported backends

1. mMiniLMv2 (117M): lightweight baseline.
2. BGE-Reranker-v2-m3 (567M): multilingual reranker.
3. Jina-Reranker-v2 (278M): multilingual long-context model.
4. Qwen3-Reranker-0.6B: causal-LM reranker in FP16.

GPU-safe execution

$$\mathrm{Load}(M_i) \rightarrow \mathrm{Run}(M_i)$$

$$\rightarrow \mathrm{Unload}(M_i) \rightarrow \mathrm{empty\_cache}()$$

Candidates: $K = 30$
Final evidence: $N = 10$

Why rerank?

Unlike separate bi-encoder vectors, a cross-encoder jointly attends to the complete query and candidate passage.

Citation-Grounded Answer Generation

Generator configuration

- Model: `gpt-4o-mini`
- Temperature: 0.0
- Maximum output: 1,024 tokens
- Context: top 10 reranked chunks

Prompt contract

- Cite governing legal provisions explicitly.
- Use only retrieved evidence.
- Return `INSUFFICIENT_CONTEXT` when evidence is inadequate.
- Format boolean, extractive, abstractive, and unanswerable outputs appropriately.

The LLM synthesizes evidence; it is not treated as the legal source.

Vietnamese Legal Corpus at a Glance

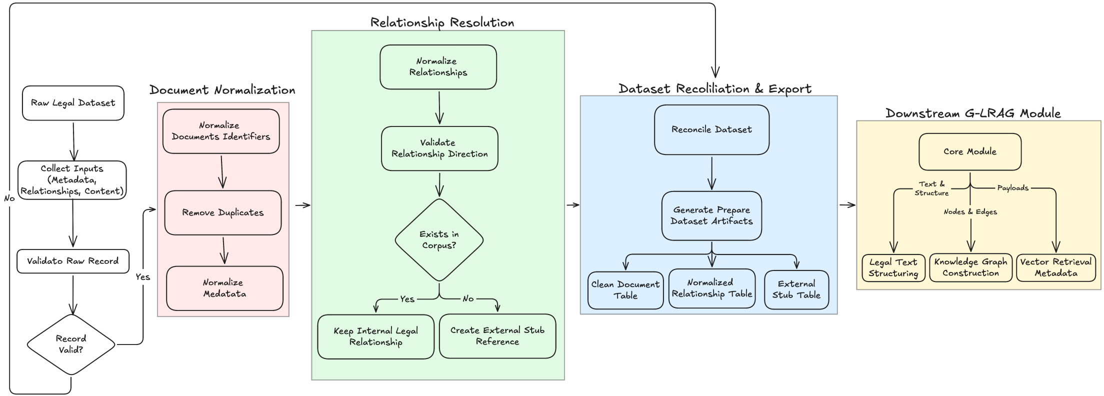
Entity	Count
Legal documents	151,624
Hierarchical provisions	1,386,267
Retrieval chunks	1,513,376
Document relationships	883,256
External referenced stubs	19,763

Corpus coverage

The collection includes National Assembly laws, Government decrees, Prime Minister decisions, and ministry circulars.

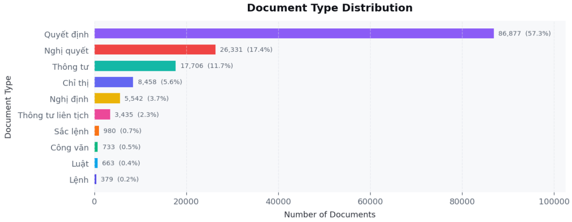
Each chunk retains document identity, provision identity, title, issuing authority, effective date, and provenance for retrieval and citation.

Corpus Preparation Pipeline

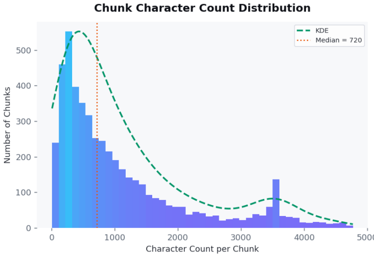


Raw metadata, relationships, and HTML content are normalized, structured into legal units, enriched with overlays, and exported for vector and graph indexing.

Document and Chunk Characteristics



Document-type distribution



Chunk-length distribution

Chunk design

The processed corpus contains 1.51M structural retrieval chunks. Chunk text is enriched with document title and provision header before embedding.

Knowledge Graph Schema

Node types

- **Document**: authority, issuer, effective date, validity.
- **Provision**: article/clause header and unit type.
- **Chunk**: retrieval text and sequence number.
- **External stub**: referenced entity outside the corpus.

Edge types

- Containment: document $\rightarrow$ provision $\rightarrow$ chunk.
- Reading order: `CHUNK_NEXT`, `PROVISION_NEXT`.
- Legal dependencies: `AMENDS`, `REPLACES`, `CITES`.
- Only verified directions may be traversed.

Integrity gating

Quarantined records and unverified edge directions are excluded from trusted traversal to reduce graph poisoning.

500-Query Legal QA Benchmark

Answerability

- 400 answerable questions
- 100 unanswerable questions
- Unanswerable cases test hallucination resistance and abstention.

Answer types

- Abstractive
- Boolean
- Extractive
- Unanswerable

Question categories

- Citation
- Cross-document
- Legal validity
- Multi-hop
- Single-hop

Difficulty levels: easy, medium, and hard.

Ground Truth and Evaluation Contract

For each answerable question, the benchmark stores:

$$Q_i = \{\text{qa\_id},\ \text{question},\ \text{category},\ \text{difficulty},\ \text{answer\_type},\ \text{ref\_answer},\ \text{gold\_chunks}\}$$

Retrieval metrics

- Recall@k and Hit@k
- MRR@k and nDCG@k
- Precision@k / chunk overlap

Generation metrics

- Exact Match
- Token F1 and ROUGE-L
- Unanswerable accuracy
- Citation grounding and faithfulness

Gold chunk IDs connect each benchmark answer directly to retrievable legal evidence.

System Status and Remaining Limitations

Delivered

- Full normalized and structured corpus.
- Dense, sparse, graph, and reranking modules.
- Query-time validity and authority overlays.
- Citation-grounded generation contract.
- Frozen 500-query benchmark.
- 27/27 graph tests passing.

Known limitations

- In-memory graph requires several GB of RAM.
- Neo4j persistence is not yet implemented.
- Some raw Vietnamese relation labels need canonical mapping.
- Backward traversal steps need clearer direction markers.
- Outputs remain a research prototype, not legal advice.

Conclusion

What L_RAG contributes

L_RAG combines semantic retrieval, exact lexical matching, explicit legal relationships, multilingual reranking, and evidence-constrained generation in one modular Vietnamese legal QA architecture.

- Metadata-enriched chunks preserve traceability from answer to provision and document.
- Dynamic graph overlays model legal authority and validity at query time.
- Sequential reranker loading supports practical multi-model experimentation on limited GPUs.
- The benchmark tests both evidence retrieval and hallucination resistance.

The system is a foundation for reliable legal retrieval, with human verification still required for real legal decisions.

Sources

- `docs/report/main.tex`: final report entry point and system summary.
- `docs/report/sections/`: introduction, related work, system overview, dataset and benchmark.
- `docs/report/FINAL_REPORT.md`: final knowledge-graph integration status and validation.
- L_RAG system, dataset, graph, retrieval, and benchmark specifications.

Questions & Discussion

Thank you!